Abstract

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

fontspec FreeSerif FreeSans

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let 1 = $((TP@title.north) - (TP@title.south))$ in node ;

(topleft) at (); (topright) at (); (lefttoright) at (); (head) at (0,); [draw=none, bottom color=blocktitlebgcolor!90!black, top color=titlebgcolor!90] (topleft) – $((topleft) - 0.2 * (head))$ .. controls $((topleft) - (head) + (1, 0))$ .. $((topleft) - (head) + 0.2 * (lefttoright))$ – $((topleft) - (head) + 0.8 * (lefttoright))$ .. controls $((topright) - (head) - (1, 0))$ .. $((topright) - 0.2 * (head))$ – (topright) – cycle; [draw=none, right color=white, left color=blocktitlebgcolor] $((topleft) - (head) + 0.8 * (lefttoright))$ .. controls $((topleft) - (head) + 0.92 * (lefttoright) + (0, 0.7))$ and $((topright) - 0.13 * (lefttoright))$ .. (topright) .. controls $((topright) - 0.11 * (lefttoright))$ and$((topleft) - (head) + 0.95 * (lefttoright))$ .. $((topleft) - (head) + 0.82 * (lefttoright))$; [draw=none, right color=white, left color=blocktitlebgcolor] $((topleft) - (head) + 0.83 * (lefttoright))$ .. controls $((topleft) - (head) + 0.97 * (lefttoright) + (0, 0.2))$ and $((topright) - 0.08 * (lefttoright))$ .. (topright) .. controls $((topright) - 0.06 * (lefttoright))$ and $((topleft) - (head) + 0.99 * (lefttoright))$ .. $((topleft) - (head) + 0.85 * (lefttoright))$;

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Fig. 1: Figures in tikzposter

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